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1. The design focuses on landfill processing 5000 tons/day of MSW for 25 years, reflecting conditions of major metropolitan areas like Mumbai.

Assumptions -

Waste generation density compacted	0.85 tons/m ³	(Manual on MSW)
Working Days	365 days/year	(Assumed)
Aug. lift height	1.0 m	(CPHEEO (2016))
Final cover slope	1V:3H	(USEPA (2000))
Daily cover	0.15 m	(Tchobanoglous et al, 2014)
Intermediate cover	0.3 m	(CPHEEO (2016))
Final cover thickness	1 m	(USEPA (2000))

$$\rightarrow \text{Annual waste Input} = 5000 \times 365$$

$$= 18,25,000 \text{ tons/yr}$$

$$\text{Total waste (25 years)} = 45,62,500 \text{ tons}$$

$$\text{Volume of compacted waste} = \frac{\text{Total waste}}{\text{Compacted Density}}$$

$$= \frac{45,62,500}{0.85}$$

$$= 53.68 \times 10^6 \text{ m}^3$$

$$\text{Adding 15\% for cover (assumed), } = 53.68 \times 1.15 \text{ million m}^3$$

$$= 61.732 \text{ million m}^3$$

Adopting a final landfill height based on metropolitan conditions to be 50 m, while maintaining manageable geotechnical stability with reinforcement.

$$\text{Required Area} = \frac{59.61.732 \times 10^6}{50} = \frac{1.235 \times 10^6 \text{ m}^2}{10^4 \text{ m}^2/\text{ha}} \\ \approx 123.5 \text{ ha}$$

→ Working Face design - Daily waste volume = $\frac{5000}{0.85}$

$$= 5882.35 \text{ m}^3/\text{day}$$

→ Assuming 3m/day face lift height and 30 m working width.

$$\text{Length} = \frac{5882.35}{3 \times 30} = 65.36 \text{ m} \approx 65 \text{ m}$$

Active cell per day → 3 m high × 30 m wide × 65 m long

Each cell covered daily with 0.15 m of soil

→ Liner System → composite liner ⇒ 60 cm compacted clay ($k \leq 1 \times 10^{-7} \text{ cm/s}$)

⇒ 1.5 mm HDPE geomembrane

⇒ 30 cm leachate drainage layer with perforated HDPE pipes.

→ Leachate Generation ~~System~~ Estimation -

Based on Tchobanoglous et al., the leachate generation rate varies 0.1 to 0.2 m³/tonne. of waste.

$$\text{Annual leachate generation} = 0.15 \times 5000 \times 365 \\ = 273,750 \text{ m}^3/\text{yr.}$$

Based on Rainfall data, avg. Mumbai rainfall = 2.2 m/yr

Assuming runoff coeff. for the catchment area = ~~0.7~~ 0.7

$$\text{Leachate} = 2.2 \times (1 - 0.7) \times 1.235 \times 10^6 \\ = 8,15,100 \text{ m}^3/\text{year}$$

→ Design of leachate collection based on rainfall leachate generation.

Min. pipe diameter = 200 mm perforated HDPE with slope 2-3%, drainage to sump.

→ Methane yield $\approx 100 - 150 \text{ m}^3 \text{ CH}_4/\text{ton waste}$ (IPCC, 2006)

$$\begin{aligned} \text{Power Potential } (0.36 \text{ kWh}/\text{m}^3 \text{ CH}_4) &= 100 \times 45.625 \times 10^6 \times 0.36 \\ &= 4.56 \times 10^9 \text{ m}^3 \times 0.36 \text{ kWh}/\text{m}^3 \\ &= 1.64 \times 10^9 \approx 1640 \text{ GWh} \end{aligned}$$

→ Environmental Protection Measures should be implemented i.e. peripheral drains at 2% slope, Leak detection layer below liner, Gas flaring or energy recovery, 15 year post closure care (monitoring and vegetation).

2 - Deonar dumpsite in Mumbai has an estimated legacy waste of about 18.5 million tonnes based on various sources of CPCB 2022 & NEERI 2021. From landfill mining,

Composition	% by wt.	Description	Reference
Combustible (Plastics, Paper, RDF, Organics, textiles)	35	Calorific Fraction suitable for WtE	NEERI (2021), IIT-B (2020)
Recyclables (metals, glass, HDPE)	10	Recoverable via material recovery facilities	CPCB (2022)
Inerts (soil, stones, C & D waste)	55	Can be reused as cover / fill	CPHEED (2016)

Assuming 16 million tonnes of mineable legacy waste

Component	Annual Recovery (over 25 years)	Total (tonnes) (million)
Combustible	0.373 0.576 million ton/yr	14.6 5.6
Recoverables	0.1067 million ton/yr	2.7 1.6
Inert	0.506 million ton/yr	12.8 8.8

→ Combustible Fraction → RDF / WtE

calorific value : 2500 - 3000 kcal/kg

$$\begin{aligned} \text{Energy potential} &\approx 2500 \times 4.184 \times 5.6 \times 10^6 \\ &\approx 58.576 \times 10^{15} \text{ J} \approx 16,271 \text{ GWh} \end{aligned}$$

→ Recyclable Fraction → Material Recovery -

$$\text{Recyclable value } (\text{₹ } 3000/\text{ton avg}) \times 1.6 \times 10^6 \\ = \text{₹ } 480 \text{ crore}$$

→ Inert Fraction - Reuse as Landfill cover / Construction Fill.

$$\text{Volume} = \frac{8.8 \times 10^6}{1.6} \approx 5.5 \times 10^6 \text{ m}^3 \text{ of inert soil.}$$

(Assuming 1.6 t/m^3 based on CPHEEO, 2016) compaction data.

→ Benefits of Mining - Reduction of methane hotspots and fire risks, recovery of valuable land, elimination of leachate seepage and odour, Reduction in GHG emissions

3-a) For WtE scenario, the Energy potential found out to be 16,271 GWh over 15 years,,

⇒ 1085 (approx.) GWh thermal energy / year

Assuming 30% conversion efficiency,

$$\text{Electricity potential} = 325.5 \text{ GWh / year}$$

We can install mass burn incinerators or RDF based combustion system with a capacity of about 1500 ton/day plant and auxiliary systems like flue gas cleaning, heat recovery boiler, etc. based on the feasibility and revenue generation and also the environmental aspects.

$$\text{Annual Revenue} \approx \text{₹ } 6/\text{kWh} \times 325.5 \times 10^6 \\ \approx \text{₹ } 195.3 \text{ crore / yr.}$$

b) For WtM scenario, Total recyclable value was ₹ 480 crore and the quantity of recoverable material will be 1.6×10^6 tons.

Virgin raw material saving: $1.5 \times 10^6 \times 1.1 \text{ t CO}_2/\text{t}$
 $= 1.76 \text{ million tons CO}_2 \text{ e avoided}$

Reduced virgin material demand $= \frac{1.6 \times 10^6}{15}$
 $= 0.107 \text{ million tonnes}$

c) For WtL scenario, $5.5 \times 10^6 \text{ m}^3$ of inert soil will be reclaimed for reuse, the reclaimed land would be

$$\text{Land} = \frac{5.5 \times 10^6}{50} = 11 \text{ ha recovered.}$$

→ This can be used as immediate cover and embankment fill or can be used in backfilling for low-lying areas.

In $\text{m}^3/\text{year} \rightarrow \frac{5.5 \times 10^6}{15} = 0.367 \times 10^6 \text{ m}^3/\text{year}$ reuse

Life extension of new landfill cells,
 $= \frac{367000}{\left(\frac{5000 \times 365}{0.85}\right)} = 0.17 \text{ yr} \approx 62 \text{ days or } \approx 2 \text{ months}$

d) Comparison — ~~Criteria~~

Criteria	WtE	WtM	WtL
O & M cost	High	Moderate	Low
Revenue	Very High	High Moderate	High Moderate
GHG Reduction	Moderate	High	Moderate
Land Reclamation	Low	Moderate	High
Public Acceptance	Moderate	High	High
Technical	High	Moderate	Low

Recommendations for Deonar site, Adopt an integrated approach prioritizing WtM supplemented by WtE and WtL. This strategy maximizes material recovery, minimizes disposal and harness residual. It also aligns with

India's Swachh Bharat Mission and Circular Economy 2030 targets.

References -

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- 5- Tchobanoglous, G., Thiesen, H. and Vigil, S. (2014). Integrated Solid Waste Management. Engineering Principles and Management Issues McGraw Hill.
- 6 - USEPA (2000), Solid Waste Disposal Facility Criteria; Technical Manual EPA/ 530-R-93-017